Effect of External and Internal Loads on Tension Loss of Tendon-Driven Continuum Manipulators

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***Abstract*—We present a novel analytical modeling approach to investigate the combined effects of external loads (i.e., contact and gravitational loads) and internal distributed friction forces on tension loss of a generic variable-curvature tendon-driven continuum manipulator (TD-CM). Unlike previous studies, in our proposed approach, we do not assume the deformation behavior (e.g., bending shape/curvature) and friction distribution profile are known *a priori* and/or can be experimentally obtained/tuned before or during the experiments. Instead, we only assume the tendons' tensions at the base of manipulator and magnitudes and locations of external forces are available while both deformation behavior and internal distributed forces are unknown and correlated. To evaluate the efficacy of the proposed approach, we performed different sets of simulation and experimental studies in the presence of both concentrated point and distributed surface contact loads. We compared the results of the performed studies with the results obtained by a friction-less modeling technique. This comparison clearly demonstrates the phenomenal performance of the proposed modeling approach in modeling tendon tension loss of a TD-CM experiencing both internal and external loads.**

***Index Terms*—Medical robots and systems, tendon-driven continuum manipulator (TD-CM), external distributed surface contact load, modeling internal distributed friction force.**

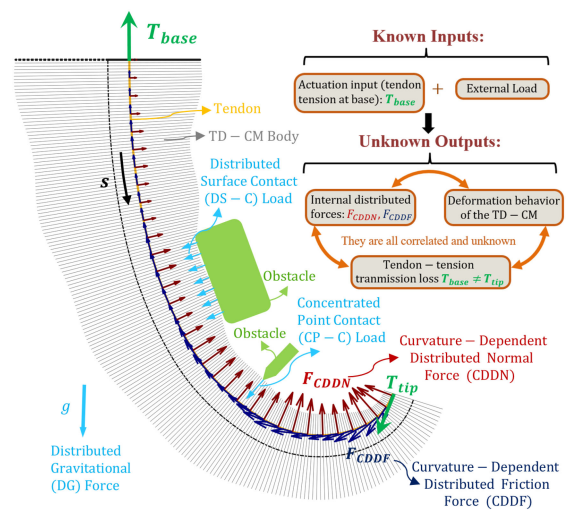

Fig. 1. Conceptual illustration of the relation between external loads, internal distributed forces, and tendon's tension loss of a generic TD-CM. Internal distributed forces, deformation behavior of the TD-CM, and tendon's tension transmission loss are all configuration-dependent, correlated, and unknown.

I. Introduction

TENDON-DRIVEN Continuum Manipulators (TD-CMs) due to their excellent dexterity, accessibility, and miniaturization capability have shown a great promise in various minimally invasive surgical (MIS) robotic applications [1], [2]. Despite the benefits of using TD-CMs for remote power transmission, they often suffer from configuration-dependent tendons' tension losses caused by internal distributed friction forces across the transmission line [1], [3]. In MIS applications, this uncertain tension loss might also be considerably amplified due to the potential contact/interaction of the manipulator with the obstructed and confined manipulator workspace within patient anatomy (Fig. 1). Of note, external point/distributed contact loads affect the deformation behavior of the manipulator, which indeed subsequently change the effect of distributed friction forces on tendons' tension losses. These effects imply the necessity of developing a mathematical model that considers the combined effects of both external contact loads and internal configuration-dependent distributed friction force on manipulator deformation behavior and tendons tension losses [1], [3]. Such comprehensive model can mitigate challenges associated with the TD-CM's design and fabrication, shape and force sensing, as well as their shape/position control [1].

Various approaches in modeling deformation behavior of TD-CMs have been presented in the previous literature in which constant-curvature (e.g., [4], [5]) and variable-curvature (e.g., [6], [7]) models are the two most popular groups. However, most of these approaches either ignored the effects of external contact loads or internal tendons' friction forces on deformation behavior of TD-CMs. To address these limitations, in this study, we will propose a deformation behavior modeling approach for a generic TD-CM that considers the effect of both external contact loads and tendons tension losses due to internal tendons' friction forces.

Several approaches have been proposed in the literature to model the effect of friction force on tendon tension loss of continuum manipulators (CMs) and tendon-sheath transmission systems. For instance, Kaneko *et al.* [8], [9] used capstan equation to model tendon-tension transmission loss and friction distribution of a fixed constant-curvature sheath configuration.

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Agrawal *et al.* [10] employed a time-domain discretized model to analyze the tendon-tension and motion transmission characteristics based on a conduit with predefined curvature. Jung *et al.* [11] proposed a nonlinear friction model that can predict the tendon tension distribution in a TD-CM. In this study, to estimate the friction parameters, authors used a 3D tracking system to capture the CM's shape followed by an off-line least squares optimization. Recently, Roy *et al.* [12] used an optimization-based method to estimate the parameters of friction and deformation of a push-pull backbone continuum manipulator. Similar to the previous study, this optimization-based method relies on the experimentally-measured data to obtain the deformation behavior and friction parameters. Overall, the proposed approaches in the literature are mainly valid for predefined, constant-curvature, or fixed sheath configurations (e.g., [9]). Additionally, they can mainly be used when deformation behavior of the robot or sheath is assumed to be known *a priori* or can be experimentally measured or estimated before the modeling procedure. More importantly, most of these friction modeling approaches have ignored the effect of external contact loads on the internal friction forces and tendon tension loss. However, in real surgical procedures, the robot usually produces unknown, variable-curvature or varying configurations, especially in a confined and obstructed space, which makes the use of these approaches very case-specific or infeasible.

The importance of modeling external loads on deformation behavior of TD-CMs has been indicated by several groups (e.g., [6], [13]). For instance, Gravagne *et al.* [14] applied an approach derived from *elastica* mechanics to investigate the influence of an applied end-effector force and external forces on deformation behavior of CMs. Jones *et al.* [15] proposed a Cosserat rod theory-based model to include the effect of gravitational force and external loads in their mathematical framework. They experimentally evaluated their model by exerting a point load at the tip of the manipulator. Rucker *et al.* [16] used Cosserat rod theory to model point and distributed wrench loads on concentric-tube robots. Later, these authors extended their approach to apply it on TD-CMs and included the effect of point moment and external distributed loads in their modeling framework [6]. Recently, Oliver-Butler *et al.* [17] have considered TD-CM under environmental loads to model the stiffness the manipulator in different conditions. Overall, review of these works demonstrates that most of the proposed modeling approaches have solely focused on considering the effect of external contact loads on deformation behavior of TD-CMs and ignored the effect of these loads on internal distributed friction forces and tension loss of these manipulators. More relevant literature can be found in [3].

Despite previous efforts in modeling the effects of external loads and/or internal tendon-sheath friction on deformation behavior and tensions' losses of TD-CMs, there is still a lack of comprehensive analytical model that can estimate the tendons' tensions losses in the presence of both external gravitational and contact loads as well as configuration-dependent internal distributed tendon-sheath friction forces along each tendon. Of note, the configuration-dependent distributed friction force depends on the deformation behavior of a TD-CM and vice versa [3]. Additionally, external contact loads affect the deformation behavior of a TD-CM and subsequently change the internal tendon-sheath distributed friction force. To consider these correlated parameters and as the main contributions of this paper, (1) we extend our previous study on modeling *unknown internal forces* (i.e., internal distributed configuration-dependent tendon-sheath friction and normal forces) in TD-CMs with planar (2D) deformation behaviors [3] and introduce a novel mathematical framework that can also model the effect of *known external forces* (i.e., Distributed Gravitational (DG), Concentrated Point Contact (CP-C), and Distributed Surface Contact (DS-C) loads) on the tendon-sheath internal forces, tendons' tensions losses of TD-CMs with a generic geometry during both bending and unbending motions. Unlike previous studies, our proposed approach can not only model external CP-C Loads or single line/curve distributed CLs but it also can consider the effect of DS-C loads on tendon-tension transmission loss of TD-CMs with a generic geometry. Additionally, to solve the developed model, we do not assume *a priori* known deformation behavior (e.g., bending shape or curvature) of the TD-CMs. Instead, we assume only tendon tensions input and external loads input are known and both the deformation behavior and internal distributed forces (e.g., tendon-tension distribution, friction and normal force) are correlated and unknown; (2) The proposed model is generic since (i) it is not geometry-, application-, or configuration-specific and can be extended to any form of TD-CMs with a continuous neutral axis; and (ii) it does not rely on experimentally-measured data and optimization-based calibration techniques, which are only valid in certain experimental conditions/configurations and are not scalable; (3) We evaluated the performance of the proposed model using simulation studies and a novel experimental setup including a silicon-molded TD-CM sensorized with two load cells to measure tendon's tension at the base and tip of the manipulator and in the presence of external gravitational, CP-C, and DS-C loads.

II. Problem Formulation

A. Problem Statement

As shown in Fig. 2, let's consider a variable-curvature TD-CM with a generic geometry, $i \in \{1, 2\}$ tendons, and a continuous neutral axis that has a planar (2D) deformation behavior. The goal is to establish an analytical relation between the known inputs (i.e., tendons' tension at the base of TD-CM and external gravitational, CP-C and DS-C loads) to unknown curvature-dependent internal distributed forces, tendons' tension transmission loss, and deformation behavior of a generic TD-CM. Of note, the considered unknown internal distributed forces include Curvature-Dependent Distributed Normal (CDDN) force $F_{\text{CDDN}}(s)$, and Curvature-Dependent Distributed Friction (CDDF) force $F_{\text{CDDF}}(s)$, which are all *a priori* unknown and functions of arc length s. With this objective in mind, the following remarks/assumptions are made in this study:

Remark 1: In this study, we assume the deformation behavior of the TD-CM is planar and the considered external CP-C and DS-C loads and tendons' tensions create a 2D deformation behavior for the manipulator.

Remark 2: In this study, we assume the tendons' elongation, backlash, and shear strains are negligible.

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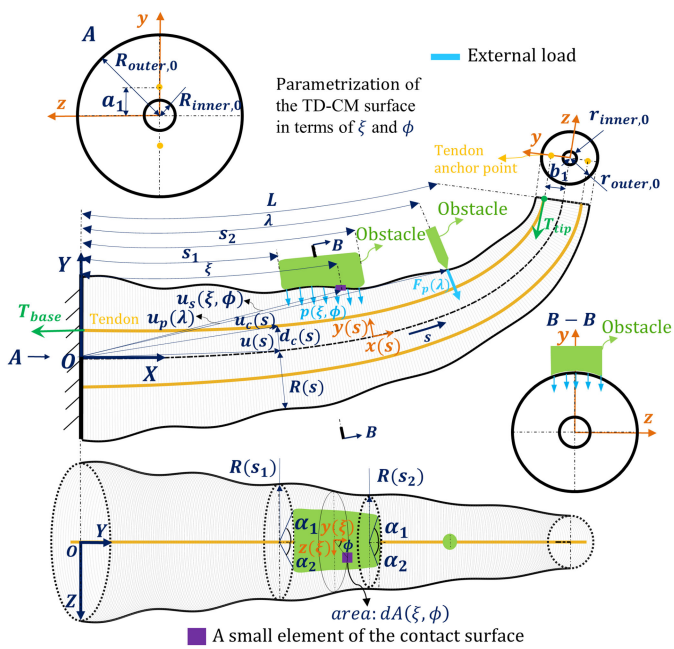

Fig. 2. Conceptual illustration of a variable-curvature TD-CM with a generic geometry that consists of $i \in \{1, 2\}$ tendons and planar deformation behavior. The geometry surface S of the manipulator can be defined using parameters ξ and ϕ, where ξ represents the arc length of the TD-CM midline from its base cross section to an arbitrary cross section and ϕ represents the angle of revolution about $\vec{t}$ axis. L denotes the overall length of TD-CM midline. λ represents the arc length of the TD-CM midline from its base cross section to a cross section on which the CP-C Load is applied. Also, $S(\xi, \phi)$ where $\xi \in [s_1, s_2]$ and $\phi \in [\alpha_1, \alpha_2]$ denotes a surface on which the DS-C Load is applied.

B. Considered Reference Frames

As shown in Fig. 2, the configuration of an arbitrary cross section in a generic TD-CM with planar deformation behavior can be described using a position vector $u(s) \in \mathbb{R}^2$ and an orientation matrix $\mathcal{R}(s) = [\vec{x}(s), \vec{y}(s)] \in SO(2)$ along the midline arc length $s \in [0, L]$. Unit vectors $\vec{y}(s) \in \mathbb{R}^2$ of the body frames lie on the plane of the cross sections, while vector $\vec{x}(s) \in \mathbb{R}^2$ is always tangent to the midline arc length. Therefore, the configuration of each body frame with respect to the global frame (i.e., $T_{gb}(s) \in SE(2)$) along the manipulator body can be represented as the following:

$$T_{gb}(s) = \begin{pmatrix} \mathcal{R}(s) & u(s) \\ 0^T & 1 \end{pmatrix} \tag{1}$$

Also, the angular and linear rate of changes of configuration $T_{gb}(s) \in SE(2)$ with respect to the arc length parameter s can be defined by the twist vector of strains $\nu_b(s) = [\omega_b(s), v_b(s)]^T \in \mathbb{R}^3$ of TD-CM body, where $\omega_b(s) = \kappa(s)$ and $v_b(s) = [1, 0]^T$ denote angular and linear strains, respectively. As shown in Fig. 2, $\kappa(s) \in \mathbb{R}$ represents the midline curvature along the arc length parameter s. Using (1), the body twist ν_b can then be obtained as follows [18]:

$$T_{gb}^{-1}(s)\frac{dT_{gb}(s)}{ds} = [\nu_b(s)] \tag{2}$$

$$\frac{dT_{gb}(s)}{ds} = \begin{pmatrix} \mathcal{R}(s) & u(s) \\ 0^T & 1 \end{pmatrix} \begin{pmatrix} [\omega_b(s)] & v_b(s) \\ 0^T & 0 \end{pmatrix} \tag{3}$$

where $[\nu_b(s)] \in se(2)$ is the matrix representation of twist vector $\nu_b(s) \in \mathbb{R}^3$ and $[\omega_b(s)] \in so(2)$ is the skew-symmetric matrix representation of angular rate of change $\omega_b(s) \in \mathbb{R}^2$. Position of each tendon $\vec{u}_{ci}(s)$ in the global frame $\{XY\}$ can be calculated using:

$$\vec{u}_{ci}(s) = \vec{u}(s) + \vec{r}_{ci}(s) \tag{4}$$

where $\vec{r}_{ci}(s)$ is the position vector representing the distance between the tendon and the midline.

To model the curvature-dependent internal distributed forces, we attached Frenet-Serret frames to each tendon inside the tendon channel. Considering these frames, directions of $F_{\text{CDDF}}(s)$ and $F_{\text{CDDN}}(s)$ can be expressed using the tangential $\vec{t}_{ci}(s)$ and normal $\vec{n}_{ci}(s)$ unit vectors of each tendon's channel curve, respectively. Since the position vector $\vec{u}_{ci}(s)$ of a tendon is defined on the global frame using (4), the mathematical relation between the Frenet-Serret frame $\{t_{ci} n_{ci} b_{ci}\}$ and the position vector $\vec{u}_{ci}(s)$ can then be obtained using the following relations [19]:

$$\begin{cases} \vec{t}_{ci}(s) = \dfrac{d\vec{u}_{ci}(s)}{ds} \\ \vec{n}_{ci}(s) = \left(\dfrac{d\vec{t}_{ci}(s)}{ds}\right) / \left\| \dfrac{d\vec{t}_{ci}(s)}{ds} \right\|_2 \\ \vec{b}_{ci}(s) = \vec{t}_{ci}(s) \times \vec{n}_{ci}(s) \end{cases} \tag{5}$$

where $\kappa_{ci}(s) = \|\frac{d\vec{t}_{ci}(s)}{ds}\|_2$ represents the curvature of each tendon along the tendon's channel and $(\ldots) \times (\ldots)$ denotes the cross product of two vectors. Of note, in our derivations, we have assumed that the midline arc length s is a good approximation of the tendon arc length s_c.

C. Modelling External Loads in a Generic TD-CM

1) Concentrated Point Contact (CP-C) Load Modeling: As shown in Fig. 2, let's first consider a general external CP-C load $\vec{F}_p(\lambda)$ applied on the surface of the manipulator with an arbitrary known magnitude and direction in the planar bending plane of the manipulator. The location $\vec{u}_p(\lambda)$ of this load on the surface of TD-CM can be expressed as:

$$\vec{u}_p(\lambda) = \vec{u}(\lambda) + R(\lambda)\vec{y}(\lambda) \tag{6}$$

where $R(\lambda)$ is the outer radius of the cross section of the TD-CM at the arc length parameter $s = \lambda$. Also, the moment $\vec{M}_{CP-C}(s)$ generated by the CP-C load along the midline arc length parameter s can be expressed as the cross product of the arm vector $\vec{r}_p(s) = \vec{u}(s) - \vec{u}_p(\lambda)$ and force vector $\vec{F}_p(\lambda)$ as the following:

$$\vec{M}_{CP-C}(s) = \begin{cases} (\vec{u}(s) - \vec{u}_p(\lambda)) \times \vec{F}_p(\lambda) & s \in [0, \lambda] \\ 0 & s \in (\lambda, L] \end{cases} \tag{7}$$

2) Distributed Surface Contact (DS-C) Load Modeling: As shown in Fig. 2, the geometry surface S of the manipulator can be defined using parameters ξ and ϕ, where ξ represents the arc length of the TD-CM midline from its base cross section to an arbitrary cross section and ϕ represents the angle of revolution about $\vec{x}$ axis. Based on this parameterization, we consider a general DS-C load/pressure $\vec{p}(\xi, \phi)$ acting on the

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contact surface $S(\xi, \phi)$, where $\xi \in [s_1, s_2]$ and $\phi \in [\alpha_1, \alpha_2]$. The location $\vec{u}_s(\xi, \phi)$ of a small element of the surface on contact can be obtained as:

$$\vec{u_s}(\xi, \phi) = \vec{u}(\xi) + R(\xi)\left(\cos(\phi)\vec{y}(\xi) + \sin(\phi)\vec{z}(\xi)\right) \tag{8}$$

Using this definition, a small element of the contact surface area $dA(\xi, \phi)$ can be computed via Pappus's theorem [20]:

$$dA(\xi, \phi) = R(\xi)(1 + R'(\xi)^2)^{\frac{1}{2}} d\xi d\phi \tag{9}$$

where $R(\xi)$ is the outer radius of the cross section of the TD-CM at the arc length parameter $s = \xi$ and $R'(s)$ is the derivative of $R(s)$ with respect to s evaluated at $s = \xi$. The relation between the DS-C load/pressure $\vec{p}(\xi, \phi)$ and the differential external force $d\vec{F}_s(\xi, \phi)$ can be expressed as:

$$\begin{aligned} d\vec{F}_s(\xi, \phi) &= \vec{p}(\xi, \phi) dA(\xi, \phi) \\ &= \vec{p}(\xi, \phi) R(\xi)(1 + R'(\xi)^2)^{\frac{1}{2}} d\xi d\phi \end{aligned} \tag{10}$$

Similarly, the moment generated by the DS-C load along the midline arc length parameter s can be expressed as the integral of the cross product of the arm vector $\vec{r}_s(s) = \vec{u}(s) - \vec{u}_s(\xi, \phi)$ and force vector $d\vec{F}_s(\xi, \phi)$, where $\xi \in [s_1, s_2]$ and $\phi \in [\alpha_1, \alpha_2]$:

$$\vec{M}_{DS-C}(s) = \begin{cases} \int_{\alpha_1}^{\alpha_2} \int_{s_1}^{s_2} (\vec{u}(s) - \vec{u}_s(\xi, \phi)) & s \in [0, s_1] \\ \quad \times \left(\vec{p}(\xi, \phi) R(\xi)(1 + R'(\xi)^2)^{\frac{1}{2}}\right) d\xi d\phi & \\ \int_{\alpha_1}^{\alpha_2} \int_{s}^{s_2} (\vec{u}(s) - \vec{u}_s(\xi, \phi)) & s \in (s_1, s_2] \\ \quad \times \left(\vec{p}(\xi, \phi) R(\xi)(1 + R'(\xi)^2)^{\frac{1}{2}}\right) d\xi d\phi & \\ 0 & s \in (s_2, L] \end{cases} \tag{11}$$

3) Distributed Gravitational (DG) Force Modeling: Let's define ρ as the density of the CM material and $\vec{g}$ as the gravitational acceleration. As shown in Fig. 2, the area of a cross section at location $s = \eta$ of the TD-CM can be defined as $A_c(\eta)$. Also, $\vec{u}_g(\eta)$ denotes the position vector of this cross section. Using these parameters, the differential gravitational force can be expressed as:

$$d\vec{G}(\eta) = \rho \vec{g} dV = \rho A_c(\eta) \vec{g} d\eta \tag{12}$$

where V is the volume of the manipulator body. Similarly, the moment $\vec{M}_{DG}(s)$ generated by the DG load along the midline arc length parameter s can be expressed as the integral of the cross product of the arm vector $\vec{r_g}(s) = \vec{u}(s) - \vec{u}_g(\eta)$ and gravitational force vector $d\vec{G}(\eta)$:

$$\vec{M}_{DG}(s) = \int_s^L (\vec{u}(s) - \vec{u}_g(\eta)) \times \rho A_c(\eta) \vec{g} d\eta \tag{13}$$

D. The Total Moment $\vec{M}_E(s)$ of the External Loads

The total moment $\vec{M}_E(s)$ due to the external CP-C load (7), DS-C load (11), and DG load (13) can then be obtained by adding these vectors:

$$\vec{M}_E(s) = \vec{M}_{CP-C}(s) + \vec{M}_{DS-C}(s) + \vec{M}_{DG}(s) \tag{14}$$

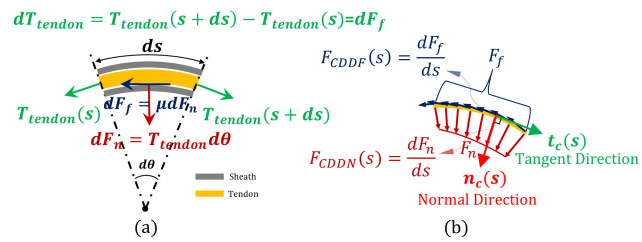

Fig. 3. (a) Forces acting on a small segment of a tendon-sheath transmission system; (b) Illustration of the relation between internal distributed forces and the Frenet-Serret frame. For an arbitrary small element, friction is in the $\vec{t_c}(s)$ direction (i.e., sliding in the tangential direction of the tendon channel) and the normal force is in the $\vec{n_c}(s)$ direction (i.e., normal to the tendon channel).

E. Modelling Distributed Friction Force in a Generic TD-CM

As illustrated in Fig. 3, for a small material element ds along the manipulator shape, the infinitely small change in each tendon tension is dT_{tendon} and its sliding direction can be defined as $\delta = -sign(T'_{tendon})$, where $T'_{tendon} = [T_{tendon}(s + ds) - T_{tendon}(s)]/ds$. Using capstan equation [9], tendon-tension distribution $\vec{T}_{tendon}(s)$ and the internal distributed forces $\vec{F}_{CDDN}(s)$ and $\vec{F}_{CDDF}(s)$ for each tendon can be expressed as functions of s [3]:

$$\begin{cases} \vec{T}_{tendon,i}(s) = -\vec{t}_{ci}(s) T_{base,i} e^{-\delta_i \mu \int_0^s \kappa_{ci}(\xi)\, d\xi} \\ \vec{F}_{CDDN,i}(s) = \vec{n}_{ci}(s) \kappa_{ci}(s) T_{base,i} e^{-\delta_i \mu \int_0^s \kappa_{ci}(\xi)\, d\xi} \\ \vec{F}_{CDDF,i}(s) = -\delta_i \vec{t}_{ci}(s) \mu \kappa_{ci}(s) T_{base,i} e^{-\delta_i \mu \int_0^s \kappa_{ci}(\xi)\, d\xi} \end{cases} \tag{15}$$

where $\theta_i(s) = \int_0^s \kappa_{ci}(\xi)\, d\xi$ denotes the wrap angle of each tendon over its channel and μ is the friction coefficient of tendons sliding along their channel. Derivation details of (15) can be found in [3].

F. Modelling the Deformation Behavior Under the External Loads and Internal Distributed Forces in a Generic TD-CM

Using (15), the internal moment $\vec{M}_{point,i}(s)$ due to the tension at the anchor point, and $\vec{F}_{CDDN}(s)$ and $\vec{F}_{CDDF}(s)$ loads can be expressed by:

$$\begin{cases} \vec{M}_{point,i}(s) = -\vec{t}_{ci}(L) T_{base,i} e^{-\delta_i \mu \theta_i(L)} \\ \quad \times [\vec{u}(s) - \vec{u}_{ci}(L)] \\ \vec{M}_{CDDN,i}(s) = \int_s^L \vec{n}_{ci}(\xi) \kappa_{ci}(\xi) T_{base,i} e^{-\delta_i \mu \int_0^s \kappa_{ci}(\xi) d\xi} \\ \quad \times [\vec{u}(s) - \vec{u}_{ci}(\xi)]\, d\xi \\ \vec{M}_{CDDF,i}(s) = -\int_s^L \delta_i \vec{t}_{ci}(\xi) \mu \kappa_{ci}(\xi) T_{base,i} \\ \quad \cdot e^{-\delta_i \mu \int_0^s \kappa_{ci}(\xi) d\xi} \times [\vec{u}(s) - \vec{u}_{ci}(\xi)]\, d\xi \end{cases} \tag{16}$$

Based on Euler-Bernoulli beam theory and equilibrium equations (14) and (16), the following equation relates the known actuation inputs, known external contact loads (i.e., gravitational, CP-C, and DS-C loads), unknown internal distributed forces (i.e., $\vec{F}_{CDDN}(s)$ and $\vec{F}_{CDDF}(s)$) to the deformation behavior

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of the TD-CM:

$$EI(s)\kappa(s) = \sum_{i=1}^{2} \Bigg\{ \int_s^L [\vec{n}_{ci}(\xi)\kappa_{ci}(\xi) - \delta_i \vec{t}_{ci}(\xi)\mu\kappa_{ci}(\xi)] \cdot T_{base,i} e^{-\delta_i \mu \int_0^s \kappa_{ci}(\xi) d\xi} \times [\vec{u}(s) - \vec{u}_{ci}(\xi)]\, d\xi - \vec{t}_{ci}(L_i) T_{base,i} e^{-\delta_i \mu \int_0^L \kappa_{ci}(\xi)\, d\xi} \times [\vec{u}(s) - \vec{u}_{ci}(L)] \Bigg\} + \vec{M}_E(s) \tag{17}$$

where E is the Young's modulus of the TD-CM and $I(s)$ defines a continuous and differentiable function for the planar second moment of area of the TD-CM's cross sections. Derivation details of (17) can be found in [3].

III. Iterative Algorithm to Solve (17)

As described in [3], to solve (17), we first ignore the effects of external loads and distributed friction forces and calculate the deformation behavior of the TD-CM in this simplified situation. The output of this step will then be used as an initial guess for iterative finding of the internal distributed forces and tendon-tension transmission loss of the robot under both internal and external loads. To terminate the algorithm, either number of iterations needs to exceed k_{max} or the following two criterion should be simultaneously satisfied:

$$\begin{cases} e_{tip}^k = \|\vec{u}_{tip}^k - \vec{u}_{tip}^{k-1}\|_2 < e_{tip,th} \\ e_{shape}^k = \sum_{l=1}^N \|\vec{u}_l^k - \vec{u}_l^{k-1}\|_2\}/N < e_{shape,th} \end{cases} \tag{18}$$

where $\vec{u}_{tip}^k$ represents the tip position of the TD-CM at the kth iteration and e_{tip}^k is the tip position increment between two successive iterations. Also, $\vec{u}_m^k$ denotes the mth cross-section position at the kth iteration and $e_{shape,th}$ is the average increment of total N calculated cross-section positions between two successive iterations. The performance and convergence of this iterative algorithm have been demonstrated in [3].

IV. Evaluation Studies

A. Simulation Studies

For the simulation studies, as shown in Fig. 2, we considered a TD-CM with a conical-shape sine-wave geometry defined by the following relation:

$$R(s) = (R_{outer,0} - L(r_{outer,0} - R_{outer,0})s) \times (1 + 0.05 \times \sin(-L \cdot 10 \cdot \pi \cdot s)) \tag{19}$$

where $L = 300$ mm, $R_{outer,0} = 30$ mm, $r_{outer,0} = 10$ mm, and $R_{inner,0} = r_{inner,0} = 7$ mm. We also assumed actuating tendon is located at $r_{c1}(s) = 9$ mm and, based on our previous work [3], used $E = 0.5$ MPa, $\rho = 1000\ kg/m^3$, and $\mu = 0.42$. Moreover, $e_{tip,th} = 0.1$ mm, $e_{shape,th} = 0.1$ mm, $N = 300$, and $k_{max} = 100$ were used as the stopping criteria for the iterative algorithm in both simulation and experimental studies.

To investigate the effect of external forces on tension loss of the described TD-CM, we performed three types of simulations with identical input tensions $T_{base,1} = 45$ N and applied the following external loading scenarios on the TD-CM: (i) CP-C, DS-C and DG loads, (ii) CP-C and DG loads, (iii) DG load only. As illustrated in Fig. 4(a), we defined CP-C and DS-C Loads using the following parameters $F_p(\lambda) = 3$ N, $\lambda = 263$ mm and $p(\xi, \phi) = 7200\ N/(m^2)$, where $\xi \in [s_1 = 100, s_2 = 150]$ mm and $\phi \in [\alpha_1 = -30°, \alpha_2 = 30°]$. Fig. 4 compares and illustrates the obtained deformation behaviors as well as the internal distributed forces (i.e., $T_{\text{CDDN}}(s)$ and $T_{\text{CDDF}}(s)$) along the tendon's arc length between these simulation scenarios. Additionally, Fig. 4(b) shows the tension loss profile along the actuating tendon and Fig. 4(c) compares the curvatures and wrap angles along arc length parameter s in these scenarios.

B. Experimental Setup

For our experiments, we used a silicone (Smooth-Sil 940, Smooth-On, Inc.) with Young's modulus of $E = 1.4\ MPa$ and density of $\rho = 1180\ kg/m^3$ to mold a TD-CM shown in Fig. 5. For this manipulator, we considered a conical geometry with the following parameters defined in Fig. 2: $L = 140$ mm, $r_{c1}(s) = 10$ mm, $R_{outer,0} = 20$ mm, $r_{outer,0} = 13$ mm, $R_{inner,0} = r_{inner,0} = 7$ mm. Then, we experimentally determined the friction coefficient ($\mu = 0.42$) between the utilized actuation tendon (wire rope: 34235T28, McMaster-Carr, USA) with 0.046 in diameter and the tendon's channel of the molded manipulator. Details of this experiment can be found in [3]. To experimentally determine the tendon tension loss under the combined effects of external gravitational load, CP-C and DS-C loads, and internal distributed friction force along the tendon, we fabricated the setups shown in Fig. 5. This setup consists of the aforementioned molded manipulator, a linear stage mechanism (M-UMR12.40, Newport Corporation), a force gauge (Mark-10 M5-20 Force Gauge, 20 LBF) to measure the external contact loads, a GemRed digital angle finder protractor to set the external loads contact angles, a tension load cell (LSB201, FUTEK Advanced Sensor Technology, Inc.) to measure the actuation tension at the base, and a through hole compression load cell (LTH300, FUTEK Advanced Sensor Technology, Inc.) mounted at the tip of manipulator to measure tendon tension loss at the tip of manipulator. Of note, the considered maximum pulling tensions in the following experiments were selected/limited based on the maximum loading capacity and linear displacements of the load cells and linear actuators, respectively.

C. Experimental Procedure

To perform experiments and generate 2D/planar deformation behaviors in the pulling/bending phase, we used one actuation tendon and continuously generated input tensions by pulling the tendon using the linear actuation mechanisms. During the actuation, we simultaneously measured the tensions at the base (i.e., $T_{base} = 0$ N-50 N with 5 N increments) and corresponding tension at the tip of manipulator using the utilized tension and compression load cells, respectively. At the end of the bending phase, we released the tendon and measured the tendon's tensions at the base and tip of manipulator (i.e., releasing/unbending phase). Of note, to consider the effect of distributed gravitational force on tendons' tension loss of TD-CM, in the performed experiments, the manipulator is bent against the gravitational direction. To thoroughly examine the performance of our approach in predicting tension loss of the tendon during these phases, we performed the following types of experiments:

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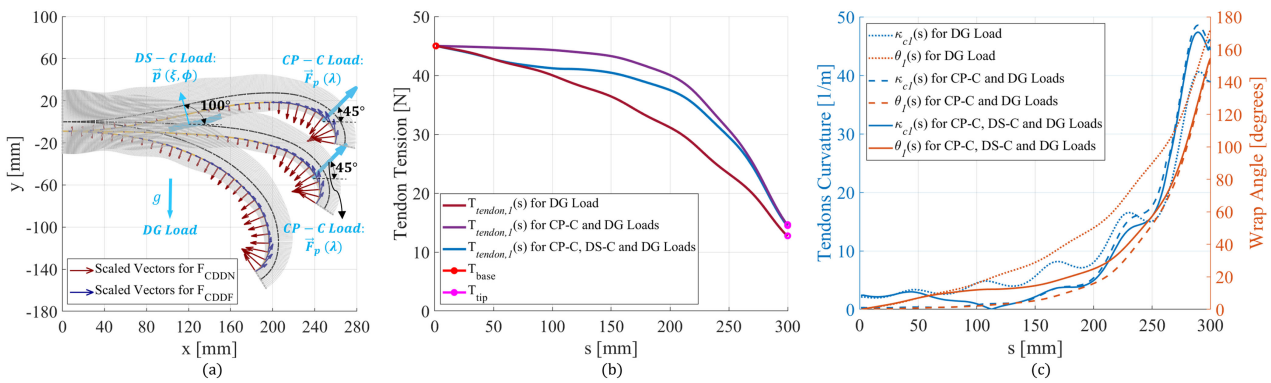

Fig. 4. Simulation results of the considered three external loading scenarios with $T_{base,1} = 45$ N: (a) Comparison and illustration of the deformation behaviors as well as the internal distributed forces (i.e., $T_{\text{CDDN}}(s)$ and $T_{\text{CDDF}}(s)$) along the TD-CM's tendon arc length between the three different external loading scenarios: (i) CP-C, DS-C and DG Loads (top configuration), (ii) CP-C and DG Loads (middle configuration), and (iii) DG Load only (bottom configuration); (b) Comparison of tension losses along the arc length s; and (c) Comparison of the curvatures and wrap angles along arc length s.

a) Subsequent Pulling and Releasing the Tendon (i.e., bending and unbending motions) while introducing the CP-C contact load during the course of planar motion: As shown in Fig. 5, we connected a sharp cone-point to the force gauge to continuously generate/measure external CP-C loads. To better demonstrate the effect of external contact load on tendon-tension transmission loss of the manipulator during the course of bending motion, we mounted the force gauge with 78° angle and 2 mm above the initial straight configuration of the manipulator. As shown in Fig. 5, we considered $\lambda = 60$ mm, which is the arc length distance of the TD-CM midline from its base cross section to a cross section on which the CP-C load is applied. In this mounting configuration, the manipulator first experienced a free bending motion before contacting the force gauge and then a variable external CP-C load was applied on the TD-CM. We then performed the pulling and releasing phases for three times and measured the tendon's tensions at the tip and base of manipulator as well as the external contact loads. Of note, we repeated this experiment three times. Fig. 6 compares the tendon's tensions at the manipulator tip obtained by experimental results, proposed model, and friction-less model while considering the external CP-C load and DG force during both pulling and releasing phases.

b) Subsequent Pulling and Releasing the Tendon (i.e., bending and unbending motions) while introducing the DS-C contact load during the course of planar motion: To create a DS-C load, as shown in Fig. 7, we replaced the cone-point tip of the previous experiment with a geometry defined by $S(\xi, \phi)$, where $\xi \in [s_1 = 55, s_2 = 85]$ mm and $\phi \in [\alpha_1 = -15°, \alpha_2 = 15°]$. We mounted the force gauge with 85° angle and 10 mm above the initial straight configuration of the manipulator. We then repeated a similar procedure as the previous experiment for three times. Fig. 8 represents the tendon's tensions at the manipulator tip obtained by experimental results, proposed model, and a friction-less model while considering the external DS-C load and DG force during both pulling and releasing phases.

V. Results and Discussion

As shown in Fig. 4(a) and Fig. 4(c), the conical-shape sine-wave geometry of the TD-CM creates a variable curvature undulating pattern in which the tendon's curvature is larger at the troughs and smaller in the crests of TD-CM's exterior outline. Also, as clearly observed in Fig. 4, the simulated loading scenarios influence on both internal distributed forces and tendon curvature distribution. For instance, for the case of considering only the DG load (i.e., the bottom configuration in Fig. 4(a)), the TD-CM experiences the largest deformation behavior and subsequently curvature, as compared to the other cases (i.e., the middle and top configurations in Fig. 4(a)) in which external loads are acting on the opposite direction of the bending motion. This is mainly because of the fact that both the distributed friction force and tension loss depend on the tendon's curvature and their values appropriately increase with this parameter (see (15) and Fig. 4(b)). Likewise, applying CP-C and DS-C external loads clearly changes the bending curvature pattern and subsequently distributed tendon's tension along the robot geometry (Fig. 4(b)), particularly at their applied location(s). Investigation of Fig. 4(b) and Fig. 4(c) demonstrates that although the tension distribution of the three simulated cases are different along the geometry of the manipulator, the total tension loss for all simulated scenarios are almost identical (i.e., ∼30 N). As shown in Fig. 4(c) and (15), this is mainly due to relatively identical total wrap angle of TD-CM's tendon (i.e., ∼160°) at $s = 300$ mm.

As can be observed in Fig. 6 and Fig. 8, due to the change in the direction of the internal distributed friction forces and magnitude of the applied contact forces during the bending and unbending phases, two completely different tension loss behaviors can be realized. Despite these differences, results of the external CP-C and DS-C Loads experiments demonstrate outstanding performances of the proposed model in predicting the tendon's tension loss during both bending and releasing phases of these experiments.

For the CP-C Load experiments, the mean absolute percentage error (MAPE) and the mean absolute error (MAE) in prediction of tendon's tension loss at the manipulator tip by the proposed model and experimental results are 8% and 1.39 N, respectively. Similarly, for the DS-C load experiments, the calculated MAPE and MAE are 10% and 1.60 N, respectively. It is worth emphasizing that these results have been obtained without relying on *a priori* known information on the TD-CM shape, predefined fixed path, and wrap angle. We also did not use experimental

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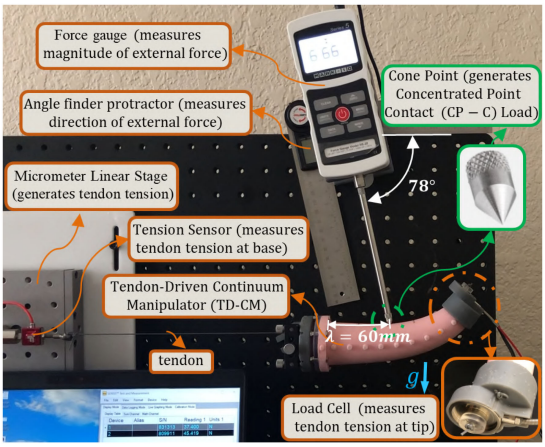

Fig. 5. Experimental setup for the CP-C Load experiment.

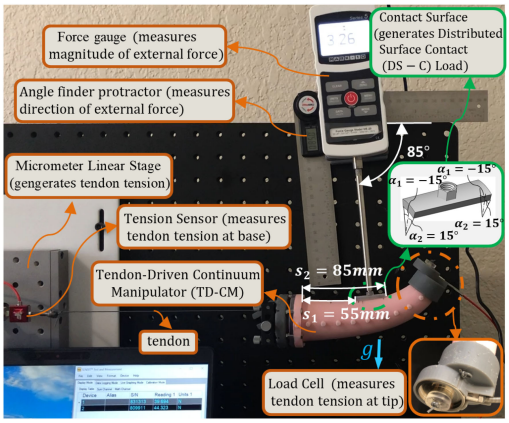

Fig. 7. Experimental setup for the DS-C Load experiment.

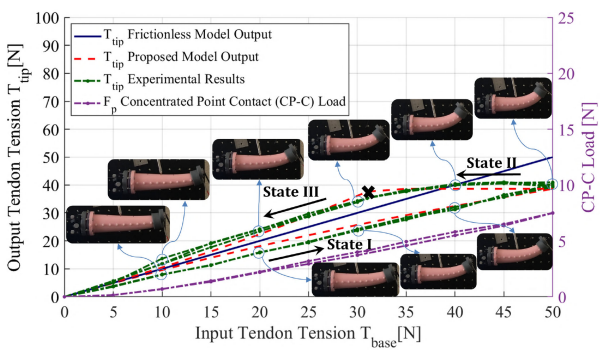

Fig. 6. Experimental results of the CP-C load experiment with the input tendon's tension ranging from $T_{base} = 0 \sim 50\,N$ with a $5\,N$ increment. The red dashed line, the blue line, and the green lines show the tendon tension at the tip of TD-CM obtained based on the proposed model, friction-less model, and experimentally-measured data from load cell, respectively. The dark purple line represents the measured/applied external loads. States I, II, and III represent *pulling*, *transition*, and *releasing* phases, respectively.

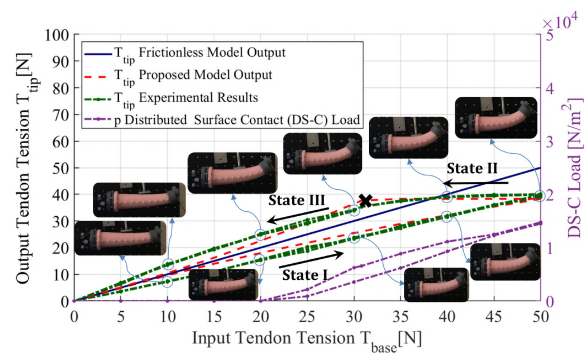

Fig. 8. Experimental results of the DS-C load experiment with the input tendon's tension ranging from $T_{base} = 0 \sim 50\,N$ with a $5\,N$ increment. The red dashed line, the blue line, and the green lines show the tendon tension at the tip of TD-CM obtained based on the proposed model, friction-less model, and experimentally-measured data from load cell, respectively. The dark purple line represents the measured/applied external loads. States I, II, and III represent *pulling*, *transition*, and *releasing* phases, respectively.

data to tune, optimize, or identify system parameters for a particular robot configuration. Of note, one of the main reasons for the discrepancy between the prediction and experimental results could be partly due to the inaccurate Young's modulus and/or experimentally measured friction coefficient. As shown in Fig. 6, the overall MAPE and the MAE between the CP-C Load experiments and the fricitonless model prediction were calculated as 21% and 4.64 N (with the MAPE = 24.74% and MAE = 5.18 N for only the pulling phase), respectively. Similarly, for the DS-C Load experiments in Fig. 8, the overall MAPE and the MAE were obtained as 18% and 4.20 N (with the MAPE = 27.57% and MAE = 5.45 N for only the pulling phase), respectively. Notably, the fricitonless model cannot capture the hysteresis phenomenon as well as the difference between the tension loss in the pulling and releasing phases under internal and external loads.

For the performed experiments, three distinct states are identified and labeled in Fig. 6 and Fig. 8. State I represents the tendon's *pulling phase*, which depicts an increase in tendon's tension loss due to an increasing cumulative distributed friction force. As shown in Fig. 6, when the input tension at the base reaches the maximum value ($T_{base} = 50\,N$), the output tip tension reaches a peak value of $T_{tip} = 40\,N$, where the maximum tension loss (at about 10 N) is observed. Also, at this point, both the CP-C and DS-C loads reach their maximum values, i.e., 7.5 N and 15 000 N/m^2, respectively. Likewise, State III indicates the tendon's *releasing phase* where the entire tendon is moving along its channel in a direction that is opposite to that of the *pulling phase*. Because of the sign's change of the cumulative distributed friction force, the output tension at the CM's tip becomes greater than the input tension at its base during the *releasing phase*. Between the *pulling phase* and *releasing phase*, there is a *transition phase* labeled as State II as shown in Fig. 6 and Fig. 8. This state occurs between the end of the *pulling phase* and the beginning of the *releasing phase*. Despite reducing the tensions at the base of manipulator, in the *transition phase*, the measured tension(s) at the tip of the manipulator will remain unchanged until certain input tendons' tensions are applied at the base. For example, in Fig. 6, when the input tension was reduced from 50 N to 31 N, the output tension at the tip remained unchanged (i.e., 38.6 N). However, any input tension less than 31 N resulted in a switch between State II and State III (the switching point is marked by a black "×"). Of note, such

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hysteresis phenomenon due to friction has also been observed and reported in the previous studies (e.g., [9], [10], [21]).

Fig. 6 and Fig. 8 also demonstrate the applied external CP-C and DS-C loads. As shown in these figures, during the *pulling phase* the measured external forces are zero when input tension is $\leq$ 5 N and $\leq$ 20 N for the CP-C and DS-C experiments, respectively. This is due to the free bending motion of the manipulator before contacting the obstacle (i.e., force gauge tip geometry). After the contact, the load continuously increases until the end of the *pulling phase* and decreases in the *releasing phase*. Also, similar to the tension loss behavior, the hysteresis phenomenon is also observed in the measured external contact loads graph. Both the CP-C load and DS-C load in the *releasing phases* are slightly greater than the *pulling phases*.

VI. Conclusion

In this letter, we proposed an analytical modeling approach and iterative solution algorithm that can predict tendons' tensions losses in the presence of external gravitational and contact loads as well as the configuration-dependent internal distributed friction force along each tendon of a generic TD-CM. We successfully evaluated the performance of the proposed approach by performing different sets of simulation and experimental studies in the presence of both external CP-C and DS-C loads with MAPE of $\leq$ 10% for the performed experiments. Our experiments also demonstrated the importance of modeling tensions' tension losses ($\sim$25% at 50 N input tension) due to the the internal and external forces as well as the hysteresis phenomenon. Of note, the presented model and obtained results did not rely on *a priori* known knowledge of the TD-CM configuration, predefined or fixed sheath path, and the wrap angle. Also, we did not use experimental data to tune, optimize or identify system parameters in some specific configurations. In the future, we will extend the proposed model to TD-CMs with 3D deformation behavior and a discrete neutral axis. We limited our experimental evaluations to a single-segment silicon molded TD-CM. Therefore, another potential extension of this study could be experimental evaluation of our proposed approach on a multi-segment TD-CM with a more complex irregular geometry, similar to the one shown in our simulations, experiencing both C- and S-shape deformation behaviors.

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